Status	Finished
Started	Tuesday, 22 September 2026, 3:33 PM
Completed	Tuesday, 22 September 2026, 3:40 PM
Duration	6 mins 43 secs
Grade	**10.00** out of 10.00 (**100**%)

Question **1**

Correct

Mark 2.00 out of 2.00

complete function to implement coin change making problem i.e. finding the minimum

number of coins of certain denominations that add up to given amount of money.

The only available coins are of values 1, 2, 3, 4

Input Format:

Integer input from stdin.

Output Format:

return the minimum number of coins required to meet the given target.

Example Input:

16

Output:

4

Explanation:

We need only 4 coins of value 4 each

Example Input:

25

Output:

7

Explanation:

We need 6 coins of 4 value, and 1 coin of 1 value

Answer: (penalty regime: 0 %)

Reset answer

```
def coinChange(amount):
    coins = [1, 2, 3, 4]

    # Initialize DP table with infinity for all amounts up to the target
    dp = [float('inf')] * (amount + 1)

    # Base case: 0 amount requires 0 coins
    dp[0] = 0

    # Compute minimum coins for all values from 1 to amount
    for i in range(1, amount + 1):
        for coin in coins:
            if i >= coin:
                dp[i] = min(dp[i], dp[i - coin] + 1)

    return dp[amount]
```

	Test	Expected	Got	
✓	print(coinChange(16))	4	4	✓

Passed all tests! ✓

Correct

Marks for this submission: 2.00/2.00.

Question **2**

Correct

Mark 2.00 out of 2.00

A strobogrammatic number is a number that looks the same when rotated 180 degrees (looked at upside down).

Write a program to determine if a number is strobogrammatic. The number is represented as a string.

Example 1:

Input:

69

Output:

true

Example 2:

Input:

88

Output:

true

Example 3:

Input:

962

Output:

false

Example 4:

Input:

1

Output:

true

For example:

Test	Result
`print(Strobogrammatic(69))`	`True`
`print(Strobogrammatic(962))`	`False`

Answer: (penalty regime: 0 %)

Reset answer

```
def Strobogrammatic(num):
    # Convert input to string in case an integer is passed
    s = str(num)

    # Valid strobogrammatic pairs when rotated 180 degrees
    mapping = {
        '0': '0',
        '1': '1',
        '6': '9',
        '8': '8',
        '9': '6'
    }

    left, right = 0, len(s) - 1

    while left <= right:
        # Check if the left character maps to the right character
        if s[left] not in mapping or mapping[s[left]] != s[right]:
            return False
        left += 1
```

```
        right -= 1

    return True
```

	Test	Expected	Got	
✓	print(Strobogrammatic(69))	True	True	✓
✓	print(Strobogrammatic(88))	True	True	✓
✓	print(Strobogrammatic(962))	False	False	✓

Passed all tests! ✓

Correct

Marks for this submission: 2.00/2.00.

Question **3**

Correct

Mark 2.00 out of 2.00

An e-commerce company plans to give their customers a special discount for Christmas.

They are planning to offer a flat discount. The discount value is calculated as the sum of all

the prime digits in the total bill amount.

Write an algorithm to find the discount value for the given total bill amount.

Constraints

1 <= orderValue< 10e100000

Input

The input consists of an integer orderValue, representing the total bill amount.

Output

Print an integer representing the discount value for the given total bill amount.

Example Input

578

Output

12

For example:

Test	Result
print(christmasDiscount(578))	12

Answer: (penalty regime: 0 %)

Reset answer

```
def christmasDiscount(orderValue):
    # Convert orderValue to string to handle large integers safely
    s = str(orderValue)
    prime_digits = {'2', '3', '5', '7'}

    total_discount = 0
    for char in s:
        if char in prime_digits:
            total_discount += int(char)

    return total_discount
```

	Test	Expected	Got	
✓	print(christmasDiscount(578))	12	12	✓
✓	print(christmasDiscount(57))	12	12	✓
✓	print(christmasDiscount(222))	6	6	✓
✓	print(christmasDiscount(77777))	35	35	✓

Passed all tests! ✓

Correct

Marks for this submission: 2.00/2.00.

Question **4**

Correct

Mark 2.00 out of 2.00

Write a function that returns the value of a+aa+aaa+aaaa with a given digit as the value of a.

Suppose the following input is supplied to the program:

9

Then, the output should be:

9+99+999+9999=11106

Sample Input Format:

9

Sample Output format:

11106

For example:

Test	Result
`print(Summation(8))`	`9872`

Answer: (penalty regime: 0 %)

Reset answer

```
def Summation(a):
    # Convert input to string to easily repeat the digit via string multiplication
    s = str(a)

    # Calculate terms: a, aa, aaa, aaaa and convert them back to integers to sum them up
    return int(s) + int(s * 2) + int(s * 3) + int(s * 4)
```

	Test	Expected	Got	
✓	`print(Summation(8))`	`9872`	`9872`	✓
✓	`print(Summation(10))`	`10203040`	`10203040`	✓

Passed all tests! ✓

Correct

Marks for this submission: 2.00/2.00.

Question **5**

Correct

Mark 2.00 out of 2.00

A number is considered to be ugly if its only prime factors are 2, 3 or 5.

[1, 2, 3, 4, 5, 6, 8, 9, 10, 12, 15, ...] is the sequence of ugly numbers.

Task:

complete the function which takes a number n as input and checks if it's an ugly number.

return ugly if it is ugly, else return not ugly

Hint:

An ugly number U can be expressed as: U = 2^a * 3^b * 5^c, where a, b and c are nonnegative integers.

For example:

Test	Result
`print(checkUgly(6))`	`ugly`
`print(checkUgly(21))`	`not ugly`

Answer: (penalty regime: 0 %)

Reset answer

```
def checkUgly(n):
    # Ugly numbers are strictly positive integers
    if n <= 0:
        return "not ugly"

    # Continuously divide by 2, 3, and 5 as long as n is divisible by them
    for prime in [2, 3, 5]:
        while n % prime == 0:
            n //= prime

    # If n reduces to 1, all its prime factors were 2, 3, or 5
    if n == 1:
        return "ugly"
    else:
        return "not ugly"
```

	Test	Expected	Got	
✓	`print(checkUgly(6))`	`ugly`	`ugly`	✓
✓	`print(checkUgly(21))`	`not ugly`	`not ugly`	✓

Passed all tests! ✓

Correct

Marks for this submission: 2.00/2.00.